

In addition to the online homework, please hand in the following problems:

Problem 1: Find a unit vector that points in the direction such that the rate of change of $f(x, y) = xy + y^2$ at $P(3, 2)$ is equal to 7?

Problem 2: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y, z) = 2xy - yz$ at $P(1, -1, 0)$ equals $-3^\circ\text{C}/\text{ft}$? If so, find it. If not, say why.

Problem 3: Prove the following properties of gradients:

a. $\nabla(kf) = k\nabla f$ for k constant

b. $\nabla \left(\frac{f}{g} \right) = \frac{\nabla f \cdot g - f \cdot \nabla g}{g^2}$